

READ ME FILE FOR “Incarceration and Incapacitation: Evidence from the 2006 Italian Collective Pardon”

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This file describes the data and programs used to generate the results in our analysis of the 2006 Italian collective pardon. There are four base data files that are used to generate the results. The first file named “monthly_data_0408.dta” is a STATA data file where each observation varies by province, month, and crime category with the key information being the number of criminal incidents reported for the year, month, province, and crime type. The data set covers all months in 2004 through 2008. The fields in this file are the following

reg_name - Italian region (larger geographic aggregate than province)

prov_name - Italian province name

crimeid - numeric code for crime type

crime_number - count of incidents

year – calendar year

month - calendar month

crime_name - crime description (in Italian)

idreg - numeric region id

idprov – numeric province id

The second data set titled “prisonertimeseries2.dta” contains monthly data on the national prison population. The data cover all months in 2004 through 2008. The fields in this file are

ymonthrelative - a variable measuring month relative to August 2006 (set to zero for August 2006)

prisoners - total inmate in Italian prisons

population - Italian population count (varies across years but not within)

incrate - prisoner per 100,000

month -month (numbered 1 through 12)

year – year

The third data set named “pardonsbyprovince.dta” contains province level counts of the number of inmates (by province of residence) that were pardoned between August 2006 and December 2006 as well as the number of prison inmates in each province on the eve of the pardon. The actual fields in this file are

idprov - a numeric province id

province_name - a province name

pardoned - the number of pardoned inmates between August 2006 and December 2006 who resided in the province prior to their commitment to prison

prisoner_jail – the province jail population prior to the pardon

The final file names “population.dta” contains the following three fields

idprov - numeric province id
 year – year
 population –province population

The program files make use of these four data sets and in some instances create intermediate data sets that are used across programs. There are four STATA do files that generate all of the inputs for the results in the table and one excel spreadsheet. The four program files, and what they do, are as follows

Figure1andpanelboftable2.do - this program uses the national level prison data to generate Figure 1 in the table and to estimate the prison time series regressions in panel B of table 2 as well as the post-pardon changes in prisons used as inputs in constructing the incapacitation effect estimates in Tables 3 and 5. The program contains detailed comments regarding each step in the calculations.

Table1figure2.do –This program creates a national level crime time series for all crime and for a set of crime sub-aggregated that we create from the detailed crime categories in the data set monthly_data_0408.dta. The program generates the statistics reported in Table 1, and the total crime rate time series presented in Figure 2. The file also generates the data set “monthnational0408.dta” which is used in the next program.

Table2andinputfortables3_4and5.do –This file merge monthly prison data to monthly national crime rates. The file then estimates the regression models in panel A of table 2 and in Table 4. The file also generates the changes in crime along the post-pardon dynamic adjustment path needed to generate some of the incapacitation effect estimates in Tables 3 and 5. Finally, the file generates IV estimates of the incapacitation effects using the discrete break in the prison time series in August 2006.

Table7and8.do - This file produces all of our province level results. It creates crime rates for each province and measures of average crime rates before and after the pardon for the time periods specified in the paper. The file also tests for diminishing returns to scale in incarceration. The file first reads the raw monthly data file "monthly_data_0408" into the program and creates province level crime rates. The file also adds data on the total number of pardons for each province through December 2006 as well as a variable measuring the province-level incarceration rate prior to the pardon.

The programs are written to be run in the same directory as the data sets listed at the beginning of this document. The reader should run the files in the order listed above.

Most of our key estimates are annualized estimates of the incapacitation effect of a prison year served. While for some estimates, we can estimate the incapacitation effect and standard errors using instrumental variables (in particular, the estimates using the discrete break in the prison time series with underlying models that do not correct for an AR1 error term), for most of our models we estimate the

standard errors of the annualized incapacitation effects using the delta methods. The four programs above generate all of the needed inputs for these estimates. We calculate the standard errors, annualized incapacitation effects, and perform a series of hypothesis tests in the excel file titled “incapacitation effect an standard error calculations for tables 3 and 5.xls.” The row and heading titles along with the formula for each cell should provide sufficient information for understanding the tabulations. Here however, we detail the contents of each column for clarity.

Note, each row corresponds to a crime-incapacitation effect estimate. For total crime, we present tabulations for all four model specifications and for each specification for the structural break, and using the change at 6, 12, 18 and 24 months following the pardon. For the individual crime aggregates, we focus only on the full model specification that includes year fixed effects, month fixed effects, and correction for AR(1) error term. For each row, we present the following variables/calculations (using the column headings to identify the variables).

Crime effect	either the discrete break in crime or the change measured at a specific point following the pardon
crime standard error	The standard error of the change in crime
prison effect	either the discrete break in the prison time series or the change measures at a specific point following the pardon
prison standard error	The standard errors of the change in the incarceration rate
incapacitation effect	The ratio of the change in the crime rate to the change in the incarceration rate multiplied by 12
inc effect standard error	The standard error of the annualized incapacitation effect estimated by the delta method
t-stat	T-statistic for the hypothesis of no incapacitation effect
abs(t-stat)	absolute value of the t-statistic
p-value	P-value from a two-tailed test of the null hypothesis of no incapacitation effect
Diff in estimates relative to break	Difference between the annualized incapacitation effect using the discrete break in the prison time series and the adjustment along the post-pardon path
Standard error	the standard error of this difference
t-stat	The t-statistic for this difference
abs(t-stat)	The absolute value for this t-statistic
P-value	P-value from a two-tailed test of the null hypothesis of no difference between the incapacitation effect estimate using the structural break in the prison time series and the estimate using the adjustment of the incarceration rate following the pardon.

